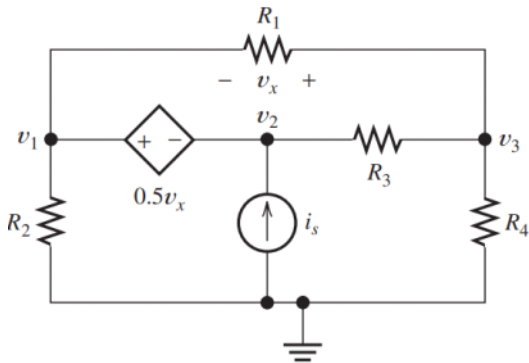
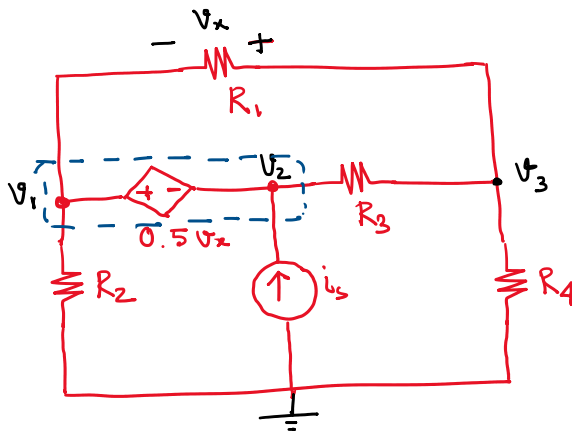
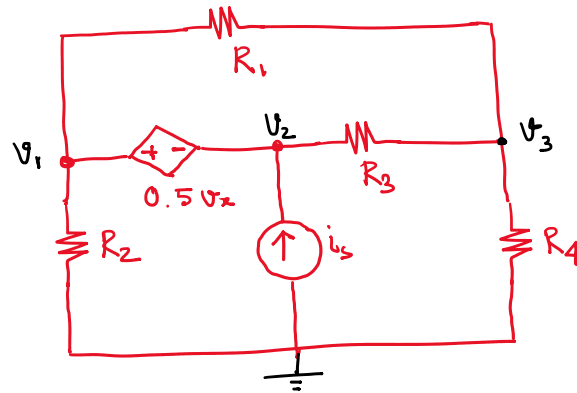


Node Analysis

Tuesday, November 19, 2024 7:55 PM



Write an independent set of equations for the node voltages shown in Figure.



① Super node 1-2:

apply KCL

$$\frac{v_1}{R_2} + \frac{v_1 - v_3}{R_1} + \frac{v_2 - v_3}{R_3} = i_s \quad \text{--- (I)}$$

$$\textcircled{2} \text{ Node 3: } \frac{v_3 - v_1}{R_1} + \frac{v_3}{R_4} + \frac{v_3 - v_2}{R_3} = 0 \quad \text{--- (II)}$$

Also, Applying KVL from node 1 to node 2

$$+v_1 - 0.5v_x - v_2 = 0$$

$$\Rightarrow v_1 - v_2 = 0.5v_x =$$

and,

Applying KVL from node 1 to node 3:

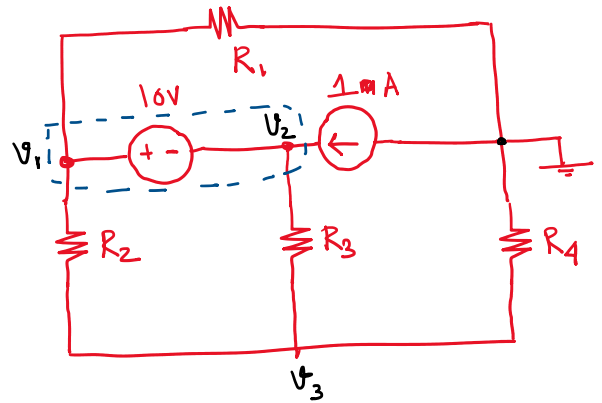
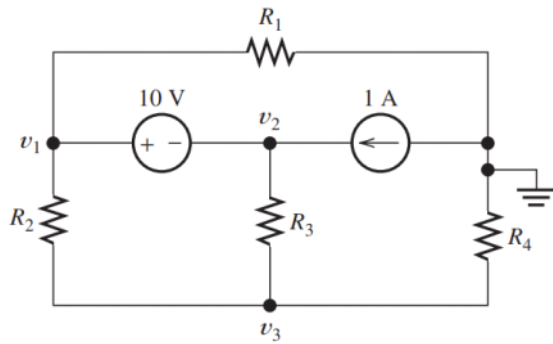
$$+v_1 + v_2 - v_3 = 0 \Rightarrow v_1 - v_3 = -v_2$$

$$\Rightarrow v_x = v_3 - v_1$$

$$\Rightarrow 0.5v_x = 0.5(v_3 - v_1)$$

$$v_1 - v_2 = 0.5v_x = 0.5(v_3 - v_1)$$

$$\Rightarrow v_1 - v_2 = 0.5v_3 - 0.5v_1 \Rightarrow 1.5v_1 - v_2 - 0.5v_3 = 0 \quad \text{--- (III)}$$



KCL at Supernode 1 & 2:

$$\frac{v_1}{R_1} + \frac{v_1 - v_3}{R_2} + \frac{v_2 - v_3}{R_3} = 1 \quad \text{--- (I)}$$

KCL at node 3:

$$\frac{v_3}{R_4} + \frac{v_3 - v_2}{R_3} + \frac{v_3 - v_1}{R_2} = 0 \quad \text{--- (II)}$$

We need the 3rd equation:

Applying KVL from node 1 to 2.

$$v_1 - 10 - v_2 = 0 \Rightarrow v_1 - v_2 = 10 \quad \text{--- (III)}$$

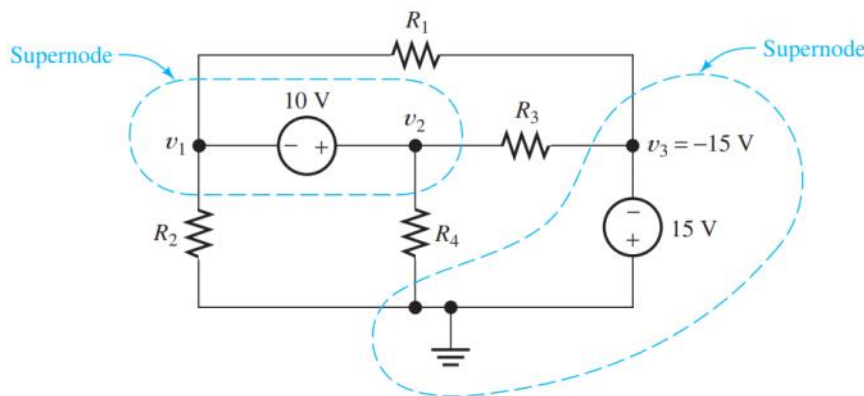


Figure 2.25 A supernode is formed by drawing a dashed line enclosing several nodes and any elements connected between them.

Applying KCL at Supernode 1 & 2:

$$\frac{v_1}{R_2} + \frac{v_1 - v_3}{R_1} + \frac{v_2}{R_4} + \frac{v_2 - v_3}{R_3} = 0 \quad \text{--- (I)}$$

Applying KVL from Ref. to node 3

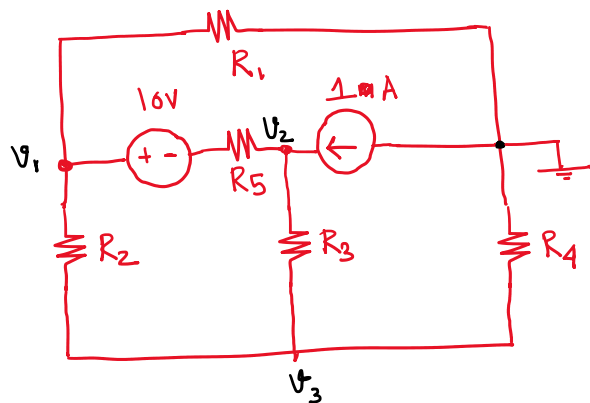
$$0 - 15 - v_3 = 0$$

$$\Rightarrow v_3 = -15V \quad \text{--- (II)}$$

Applying KVL from node 1 to 2:

$$v_1 + 10 - v_2 = 0$$

$$\Rightarrow v_2 - v_1 = 10V \quad \text{--- (III)}$$



There is a resistor R_5 between node 1 and 2, in addition to 10V source. So its not a Super node:

KCL at node 1.

$$\frac{V_1}{R_1} + \frac{V_1 - 10 - V_2}{R_5} + \frac{V_1 - V_3}{R_2} = 0 \quad \text{--- (i)}$$

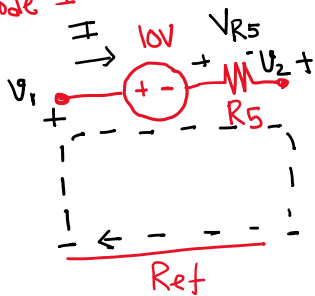
KCL @ Node 2:

$$\frac{V_2 - V_3}{R_3} + \frac{V_2 + 10 - V_1}{R_5} = 1 \quad \text{--- (ii)}$$

KCL at node 3

$$\frac{V_3}{R_4} + \frac{V_3 - V_2}{R_3} + \frac{V_3 - V_1}{R_2} = 0 \quad \text{--- (iii)}$$

@ Node 1



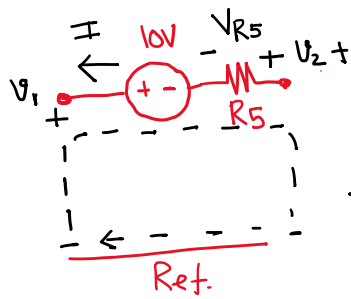
While applying KCL at node 1, unknown current is assumed to be leaving the node meaning that V_1 is the higher potential. See the current direction and corresponding

Polarity of V_{R5}

Applying KVL: $+V_1 - 10 - IR_5 - V_2 = 0$
 $\Rightarrow I = \frac{V_1 - 10 - V_2}{R_5}$

While applying KCL at node 2, unknown current is assumed to be leaving the node meaning that V_2 is the higher potential. See the current direction and corresponding
 Polarity of V_{R5}

While at N2



Applying KVL

$$+V_1 - 10 + IR_5 - V_2 = 0$$

$$\Rightarrow I = \frac{V_2 + 10 - V_1}{R_5}$$